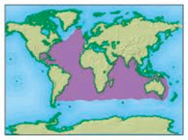
Hoplostethus atlanticus

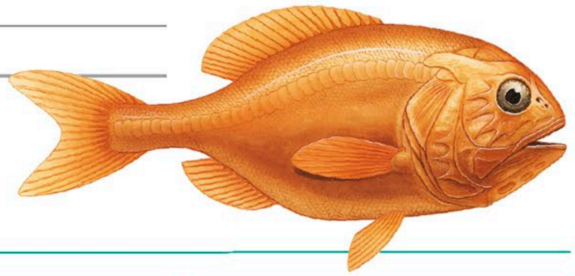
Orange roughy



Length 20–23½ in (50–60 cm)
Weight Not recorded
Sex Male/Female
Status Not evaluated

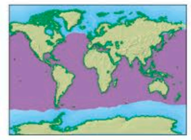
Location Atlantic, Indian Ocean, W. Pacific

Mainly found at the edges of continental shelves, or on the of the open ocean floor, the orange roughy has an upturned mouth, and small teeth arranged in bands. Its body is orange-red, although it appears dark at great depths, which gives it protection from predators. It is known to have the longest lifespan among fish, and can live up to 150 years. Like its close relative, *Hoplostethus mediterraneus*, the orange roughy is often caught as a food fish, and is very vulnerable to overfishing.



Barbourisia rufa

Velvet whalefish



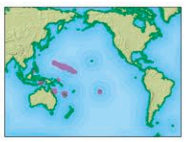
Length Up to 15½ in (39 cm)
Weight Not recorded
Sex Male/Female
Status Least concern

Location Tropical, subtropical, and temperate waters worldwide

This rarely sighted, deep-sea fish has a large mouth, the jaws having many small, depressible teeth. It is reddish orange, appearing black in deep waters, and its elongated body has a flabby feel. The skin of the velvet whalefish is also slightly prickly—a characteristic produced by its scales, each of which has a tiny, central spine. Its fins are placed far back along its body, suggesting that the fish is an ambush predator that attacks with a rapid, headlong rush. The fact that only individual specimens have been collected suggests a solitary lifestyle. This species has not been well studied, as few specimens have been collected, but their size suggests that, like many deep-sea fishes, the velvet whalefish is found at different depths at varying stages of its life. Small young fishes have been found in middle depths, whereas older fishes have been brought up from the seabed, indicating that this is where they spend their adult lives.

Photoblepharon palpebratum

Flashlight fish



Length Up to 4¾ in (12 cm)
Weight Not recorded
Sex Male/Female
Status Not evaluated

Location W. Pacific

This small, blunt-nosed fish is one of the best examples of bioluminescence—or light production—in the animal world. One of only 8 known species of flashlight fishes, it remains relatively inactive in caves and deeper waters during the day, and moves to shallower areas over reefs at night, where it feeds on small planktonic animals. Individuals defend small territories on the reef. Symbiotic bacteria, contained in an organ under the eye, produce this fish's bluish-lime light. The light helps it to find prey, startle predators, and communicate with other individuals. Using a movable black membrane to cover and expose the bacteria, the fish can signal to others in a shoal.

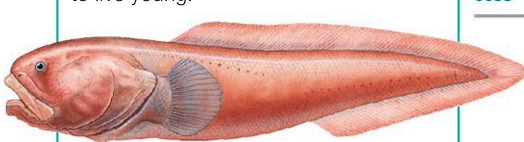
Thermichthys hollisi

Ventfish

Length Up to 12 in (30 cm)
Weight Not recorded
Sex Male/Female
Status Not evaluated

Location Galapagos Islands

First identified in 1990, this rare fish is found only around hydrothermal vents on the deep-sea bed. It has a short, stout body and a sizable mouth, but small, poorly developed eyes. Like other vent animals, it depends for its survival on the bacteria that thrive in the hot, mineral-laden water that gushes out of vents. It is not known whether it feeds on the bacteria directly or on other animals that use them for food. The ventfish belongs to a family of more than 200 species, known as live-bearing brotulas, which live both in deep-sea habitats and around shallow reefs. All give birth to live young.



Ophidion scrippsae

Basketweave cusk-eel

Length Over 11 in (28 cm)
Weight Not recorded
Sex Male/Female
Status Least concern

Location E. North Pacific

Cusk-eels are not true eels, but relatives of pearlfishes (see left) and brotulas (see ventfishes, left). The basketweave cusk-eel normally lives at depths of up to 330 ft (100 m), but some specimens have been found at over 26,400 ft (8,000 m)—the greatest depth for any fish. Like other cusk-eels, this North American species has a blunt head, large mouth, and an elongated body that tapers almost to a point. The scales on its body are arranged in a crisscross pattern, which give the species its name. Basketweave cusk-eels are active only at night; during the day, they bury themselves in sand with only the tip of their snout showing. At dusk, they partly emerge from the sand, waiting for small fishes or invertebrates to pass by. These fishes may also swim along the ocean floor in search of their food.

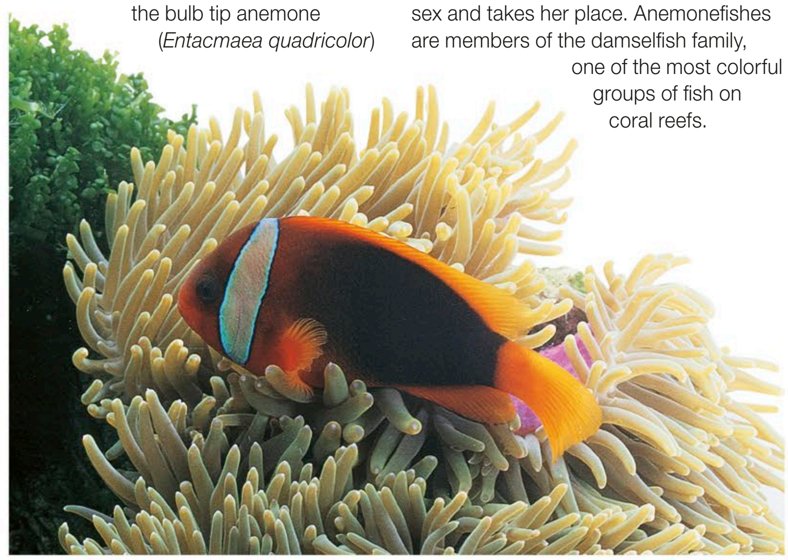
Amphiprion frenatus

Tomato clownfish

Length 3–5½ in (7.5–14 cm)
Weight Not recorded
Sex Sequential hermaphrodite
Status Not evaluated

Location W. Pacific

This small fish, like other anemonefishes, lives with sea anemones in a symbiotic relationship. It lives only with the bulb tip anemone (*Entacmaea quadricolor*) while other species may have many hosts. It protects itself from the anemone's stings by the mucus covering its body. If the tomato clownfish leaves the anemone, it must, on its return, reestablish its immunity through a series of brief contacts with the anemone, after which it can again immerse itself among the stinging tentacles without harm. It receives shelter from the anemone, and feeds on plankton floating by and algae growing around the anemone. A family unit usually occupies a single sea anemone. The largest 2 fishes on an anemone are the male and female that do all the spawning; the rest are males, one of which, when the female dies, changes sex and takes her place. Anemonefishes are members of the damselfish family, one of the most colorful groups of fish on coral reefs.



Carapus acus

Pearlfish

Length Up to 8 in (20 cm)
Weight Not recorded
Sex Male/Female
Status Not evaluated

Location E. Atlantic, Mediterranean

Like most of the other species in its family, this slim, scaleless, silvery white fish spends its adult life inside sea cucumbers. Attracted to them by their shape and the chemicals that they release, the pearlfish enters its host, tail first, through the animal's anus, and takes up residence in the body cavity. After dark, it goes out to feed on other animals, but it may also feed parasitically on its host's internal organs.